

FEDERAL BOARD OF INTERMEDIATE & SECONDARY EDUCATION

Annual Examination 2026

Chemistry — Class 9 (Science Group)

Syllabus: Chapter 1 — Nature of Chemistry | Chapter 2 — Matter | Chapter 3 — Atomic Structure | Chapter 4 — Periodic Table & Periodicity | Chapter 5 — Chemical Bonding | Chapter 6 — Stoichiometry | Chapter 7 — Electrochemistry | Chapter 8 — Energetics | Chapter 9 — Chemical Equilibrium

Total Marks: 46

Time: 2 Hours

Paper: Exam-21 (Version 1)

1. **Attempt ALL questions. There is no choice or OR alternative anywhere in this paper.**
2. Section A (30 marks): 30 MCQs. Fill the correct bubble. Suggested time: 35 minutes.
3. Show complete working for every calculation. Always write the correct unit with your final answer.
4. Use black or blue ink only. Pencil is allowed only for diagrams. Relative atomic masses are given where needed.

SECTION A — Multiple Choice Questions (30 × 1 = 30 Marks)

#	Q.1 Choose the best possible option.	A	B	C	D	Answer
1	Which branch of chemistry mainly studies the energy changes that accompany chemical reactions?	Physical chemistry	Analytical chemistry	Industrial chemistry	Nuclear chemistry	A B C D
2	Which statement best distinguishes technology from science?	Science tests a hypothesis; technology never does	Technology applies scientific knowledge to solve practical, everyday problems	Technology only studies natural phenomena	Science and technology mean exactly the same thing	A B C D
3	A mixture whose particles are too small to settle on standing, but large enough to scatter light (Tyndall effect), is called a:	True solution	Suspension	Colloid	Pure compound	A B C D
4	Which of these is an example of a solid solute dissolved in a liquid solvent to form a true solution?	Smoke in air	Muddy river water	Milk	Sugar dissolved in water	A B C D
5	Diamond, graphite and fullerenes are different physical forms of the same element, carbon. Such forms of the same element are called:	Isomers	Allotropes	Isotopes	Isobars	A B C D
6	An atom as a whole is electrically neutral because:	the number of protons equals the number of electrons	neutrons cancel the charge on the nucleus	it contains no charged particles at all	protons and neutrons repel each other equally	A B C D
7	An atom of zinc has atomic number 30 and mass number 65. The number of neutrons in this atom is:	30	65	95	35	A B C D
8	Which statement correctly describes isotopes of the same element?	Different number of protons but the same mass number	Same number of neutrons but a different number of protons	Same number of protons but a different number of neutrons	They belong to different elements entirely	A B C D
9	The Quantum Mechanical Model describes the position of an electron around the nucleus in terms of:	Probability regions called orbitals	Fixed circular orbits	A stationary point on the nucleus	Straight-line paths only	A B C D
10	Transition elements commonly show variable (more than one) oxidation state mainly because:	they belong to Group 1 of the periodic table	both their s and d sub-shell electrons can take part in bonding	their electron affinity is exactly zero	they have only one valence electron	A B C D
11	According to the structure of the periodic table, the LENGTH of a	the atomic mass of its	the melting points of its	the number of protons only,	the type of sub-shell (s, p, or d)	A B C D

	period (the number of elements in it) is determined mainly by:	elements	elements	regardless of sub-shells	being filled across that period	
12	Group 1 and Group 2 elements of the periodic table are together called the:	p-block elements	d-block elements	s-block elements	f-block elements	A B C D
13	Electron affinity is best defined as:	the minimum energy needed to remove the outermost electron from a gaseous atom	the energy released when an isolated gaseous atom gains an electron to form a uni-negative ion	the energy needed to break one mole of a covalent bond	the energy absorbed when a solid melts	A B C D
14	Which transition metal is used as the catalyst in the Haber process for manufacturing ammonia?	Iron	Nickel	Platinum	Palladium	A B C D
15	Transition elements such as iron (7.87 g/cm^3) and tungsten (19.3 g/cm^3) generally have high densities mainly because of their:	very large atomic radii	low melting points	higher atomic masses and closely packed structures	non-metallic character	A B C D
16	A covalent bond in which BOTH shared electrons are donated by only ONE of the two bonded atoms is called a:	Ionic bond	Metallic bond	Hydrogen bond	Coordinate covalent bond	A B C D
17	When potassium (electronic configuration 2, 8, 8, 1) reacts with chlorine to form potassium chloride, the potassium atom:	loses one electron to become K^+	gains one electron to become K^-	shares one pair of electrons with chlorine	loses two electrons to become K^{2+}	A B C D
18	Metals conduct electricity well mainly because they contain:	fixed ions that cannot move	a 'sea' of delocalized electrons that can move freely through the structure	only covalent bonds	no free-moving charged particles	A B C D
19	Diamond has an extremely high melting point mainly because it:	consists of small individual molecules held by weak forces	is an ionic lattice	has a giant covalent (network) structure of countless strong covalent bonds	has freely-moving delocalized electrons	A B C D
20	One mole of any substance always contains Avogadro's number of particles, which equals:	6.022×10^{22}	3.011×10^{23}	1.66×10^{-24}	6.022×10^{23}	A B C D
21	The number of moles present in 2.3 g of sodium metal is: ($\text{Na} = 23$)	0.01 mol	0.1 mol	1 mol	10 mol	A B C D
22	Which of the following has an empirical formula that is DIFFERENT from its molecular formula?	Benzene, C_6H_6 (empirical CH)	Water, H_2O	Ammonia, NH_3	Carbon monoxide, CO	A B C D
23	Balancing the decomposition of hydrogen peroxide, $__\text{H}_2\text{O}_2 \rightarrow __\text{H}_2\text{O} + __\text{O}_2$, the correct coefficients (in order) are:	1, 1, 1	2, 1, 2	1, 2, 1	2, 2, 1	A B C D
24	The oxidation number of sulfur in sulfuric acid, H_2SO_4 , is:	+4	+2	+6	-2	A B C D
25	In a redox reaction, the oxidizing agent is the substance that:	loses electrons and is itself oxidized	gains electrons and is itself reduced	remains chemically unchanged	always contains only non-metal atoms	A B C D
26	Using the rule that the sum of oxidation numbers in a polyatomic ion equals its charge, the oxidation number of chromium in the chromate ion, CrO_4^{2-} , is:	+6	+3	+4	+2	A B C D

27	A catalyst speeds up a chemical reaction mainly by:	raising the temperature of the reactants	increasing the overall enthalpy change of the reaction	providing an alternative reaction pathway with lower activation energy	being permanently used up in the reaction	A B C D
28	In an exothermic reaction, the enthalpy change (ΔH) is:	always exactly zero	positive, because energy is absorbed	impossible to measure	negative, because energy is released to the surroundings	A B C D
29	A reaction that can proceed in both the forward and reverse directions under the same conditions, and never goes fully to completion, is called a:	Reversible reaction	Complete reaction	Precipitation reaction	Neutralization reaction	A B C D
30	When hydrated cobalt(II) chloride, $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$, is heated and loses its water of crystallization, its colour changes from:	blue to pink	pink to blue	white to blue	green to colourless	A B C D

SECTION B — Short Questions ($2 \times 4 = 8$ Marks)

Attempt ALL questions. There is no choice or OR alternative anywhere in this section.

Part	Question	Marks
2(i)	Write a balanced chemical equation, with correct state symbols, for the burning (complete combustion) of hydrogen gas in oxygen to produce liquid water. State whether this reaction is exothermic or endothermic, giving a reason in terms of bond breaking and bond forming.	4
2(ii)	Aluminium ions carry a charge of +3 (Al^{3+}) and sulfate ions carry a charge of -2 (SO_4^{2-}). Using the criss-cross (valency) method, derive the chemical formula of aluminium sulfate. Show your working clearly, including the LCM step, and write the final formula with the correct bracket notation.	4

SECTION C — Long Question ($1 \times 8 = 8$ Marks)

Attempt this question. There is no choice or OR alternative.

Part	Question	Marks
3	A dicarboxylic acid, compound Z, was analysed and found to contain 49.3% carbon, 6.8% hydrogen and 43.9% oxygen by mass. Its relative molecular mass is 146. ($\text{C} = 12$, $\text{H} = 1$, $\text{O} = 16$, $N_A = 6.022 \times 10^{23}$) (a) Calculate the empirical formula of Z. [3] (b) Hence determine the molecular formula of Z. [1] (c) Calculate the number of moles of Z present in a 43.8 g sample. [1] (d) Calculate the number of molecules of Z present in this sample. [1] (e) Write a balanced equation, with correct coefficients, for the complete combustion of Z (using its molecular formula) with oxygen to form carbon dioxide and water. [1] (f) Using this balanced equation, calculate the mass of carbon dioxide produced when 0.3 mol of Z burns completely. [1]	8

Student's Signature: _____

Invigilator: _____

Date: _____

Chapters Covered: 1 – Nature of Chemistry, 2 – Matter, 3 – Atomic Structure, 4 – Periodic Table & Periodicity, 5 – Chemical Bonding, 6 – Stoichiometry, 7 – Electrochemistry, 8 – Energetics, 9 – Chemical Equilibrium

— End of Question Paper —